



Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions

Dataset Summary

3900

Rows

Total purchases
analyzed

18

Columns

Data attributes

37

Missing Values

In Review Rating
column



Data Preparation Process

1

Data Loading

Imported dataset using pandas

2

Initial Exploration

Structure check and summary statistics

3

Missing Data Handling

Imputed Review Rating with median values

4

Feature Engineerin

Created age groups and purchase frequency

5

Database Integration

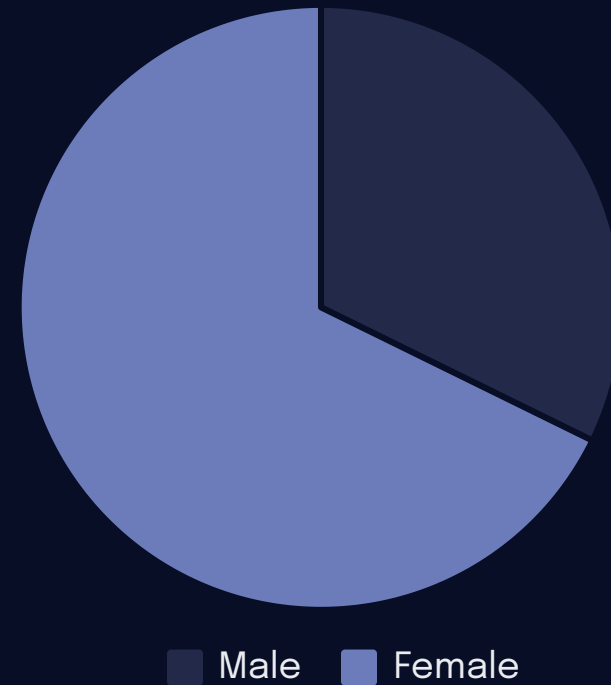
Connected to PostgreSQL for analysis

Revenue by Gender Analysis

Key Insight

Female customers generate slightly higher total revenue than male customers

- Gender-based marketing strategies could optimize revenue streams





High-Value Discount Users

Customers using discounts while spending above average



Smart Shoppers

High spenders who maximize value with discounts



Strategic Opportunities

Target premium customers with exclusive offers



Top-Rated Products

★★★★★ 5

Blouse

Highest customer satisfaction

★★★★★ 5

Dress

Consistently excellent reviews

★★★★☆ 4

Shirt

Strong customer approval

Shipping Preferences Impact

Express Shipping

\$65

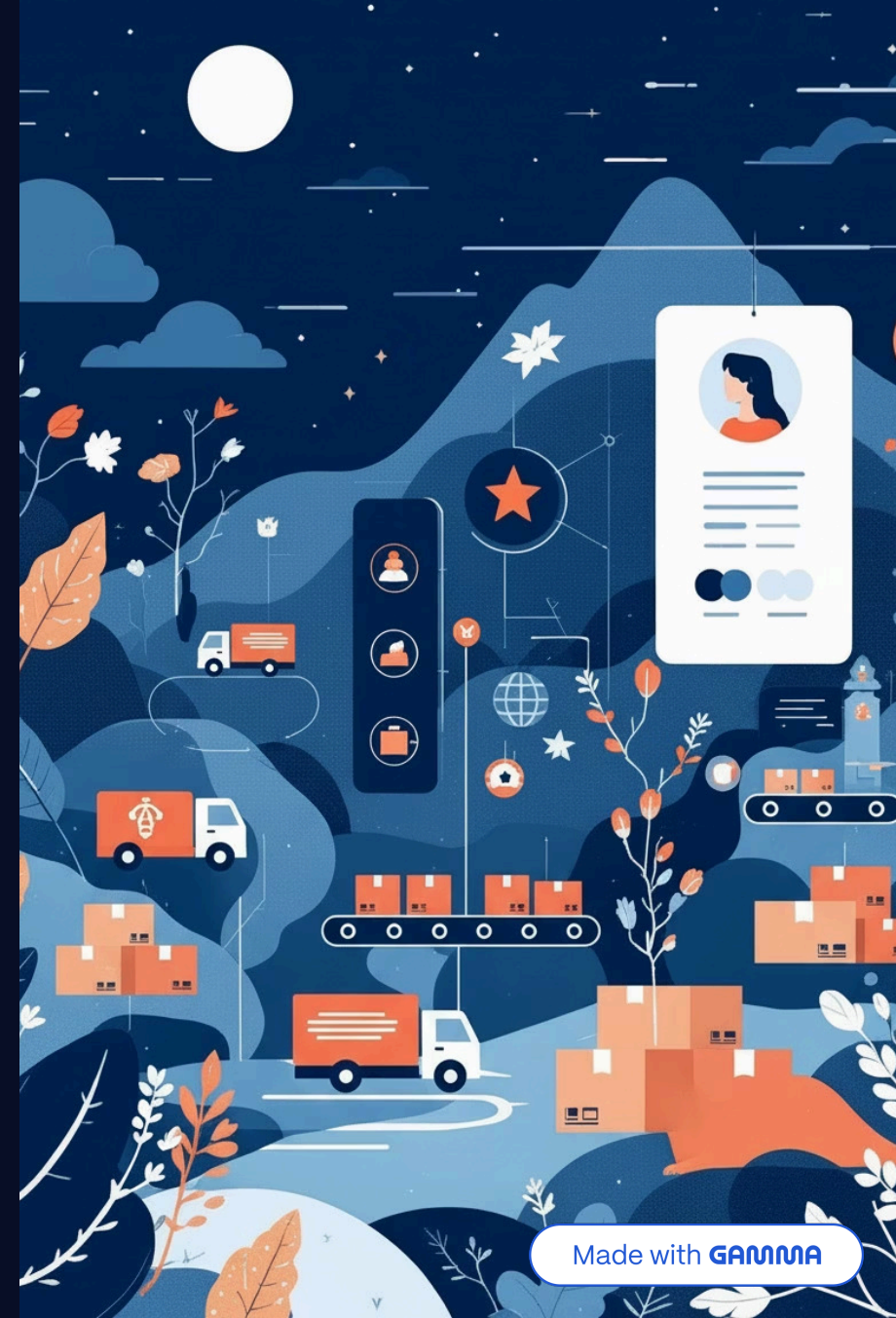
Average purchase amount

Express shipping customers spend 12% more per transaction

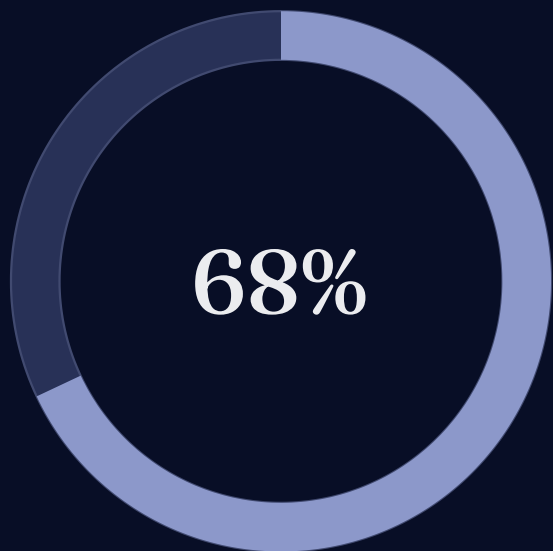
Top 3 Products per Category

\$58

Average purchase amount

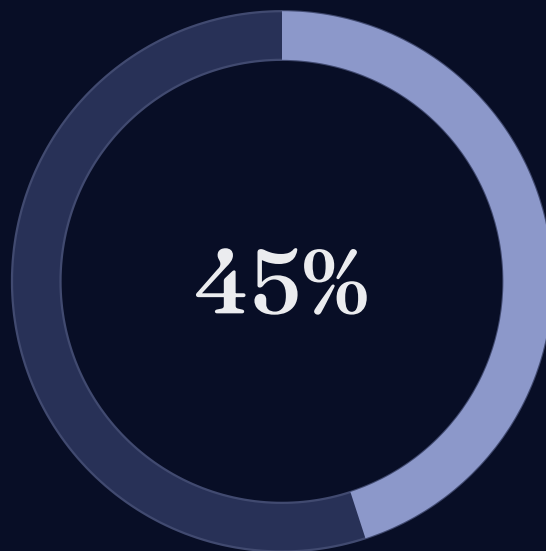


Subscription Impact



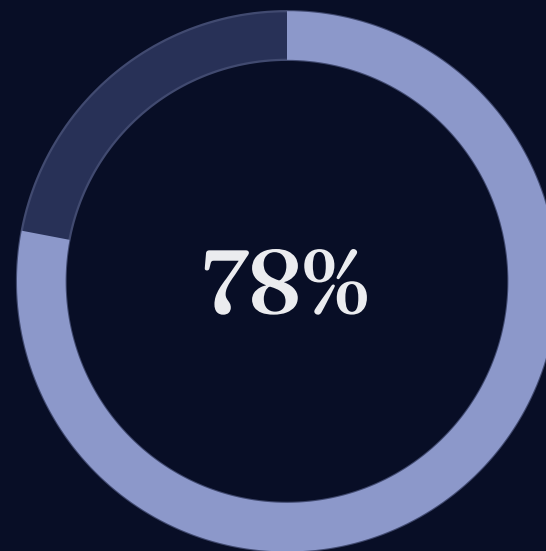
Higher Spend

Subscribers vs. non-subscribers



Revenue Share

From subscription customers



Loyalty Rate

Repeat purchase frequency

Customer Segmentation



Focus on converting New to Returning, Returning to Loyal

Business Recommendations

- Focus on improving subscriber retention and purchase frequency, rather than transaction value alone

- Convert non-subscribers into subscribers through long-term value proportions

- Apply targeted marketing to high-value customer segments